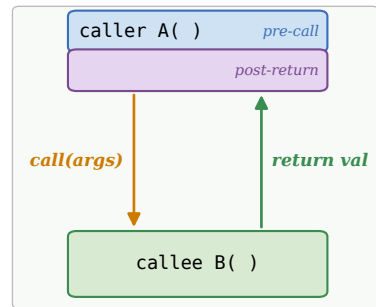
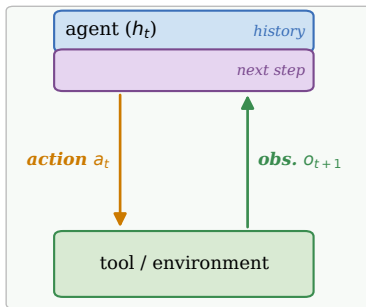


Function call site $\sim$ agent step (structurally similar) $\Rightarrow$ use as self-supervised FIM mid-training prior

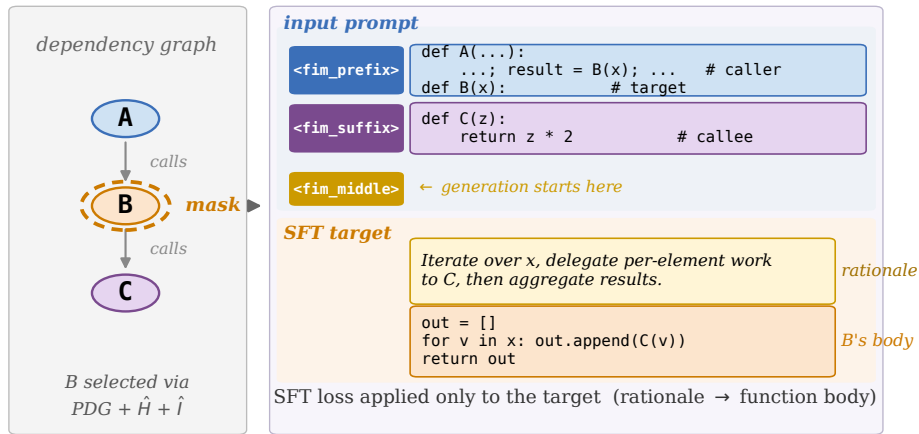
Function call site



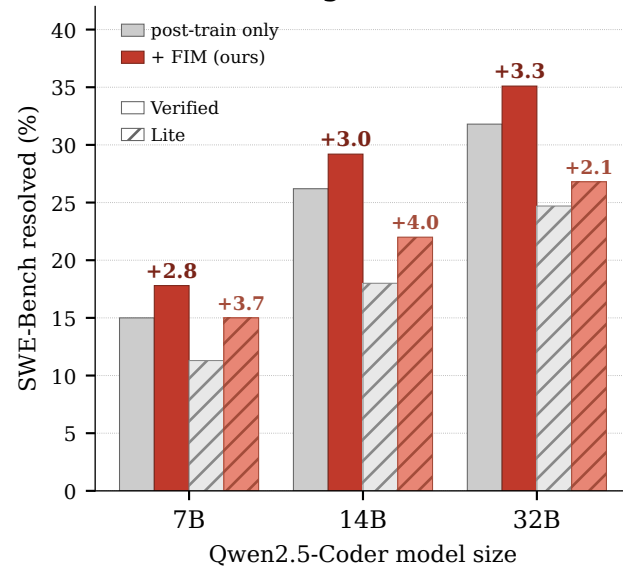
Coding-agent step



Function-level FIM mid-training (SFT)



SWE-Bench gain across scales



■ context
 ■ call / action
 ■ return / observation
 ■ continuation
 same 4-stage decomposition: context $\rightarrow$ call $\rightarrow$ return $\rightarrow$ continuation